

DNA REPAIR MECHANISM

INTRODUCTION:

Damage to DNA caused by replication errors or mutations may have serious consequences. The cell possesses an inbuilt system to repair the damaged DNA. This may be achieved by four distinct mechanisms (Table 24.2).

- 1) Base excision-repair
- 2) Nucleotide excision-repair
- 3) Mismatch repair
- 4) Double-strand break repair.

TABLE 24.2 Major mechanisms of DNA repair

<i>Mechanism</i>	<i>Damage to DNA</i>	<i>DNA repair</i>
Base excision-repair	Damage to a single base due to spontaneous alteration or by chemical or radiation means.	Removal of the base by N-glycosylase; abasic sugar removal, replacement.
Nucleotide excision-repair	Damage to a segment of DNA by spontaneous, chemical or radiation means.	Removal of the DNA fragment (~ 30 nt length) and replacement.
Mismatch repair	Damage due to copying errors (1-5 base unpaired loops).	Removal of the strand (by exonuclease digestion) and replacement.
Double-strand break repair	Damage caused by ionizing radiations, free radicals, chemotherapy etc.	Unwinding, alignment and ligation.

1) BASE EXCISION-REPAIR:

The bases cytosine, adenine and guanine can undergo spontaneous depurination to respectively form uracil, hypoxanthine and xanthine. These altered bases do not exist in the normal DNA, and therefore need to be removed. This is carried out by base excision repair (Fig.24.16).

A defective DNA in which cytosine is deaminated to uracil is acted upon by the enzyme uracil DNA glycosylase. This results in the removal of the defective base uracil. An endonuclease cuts the backbone of DNA strand near the defect and

removes a few bases. The gap so created is filled up by the action of repair DNA polymerase and DNA ligase.

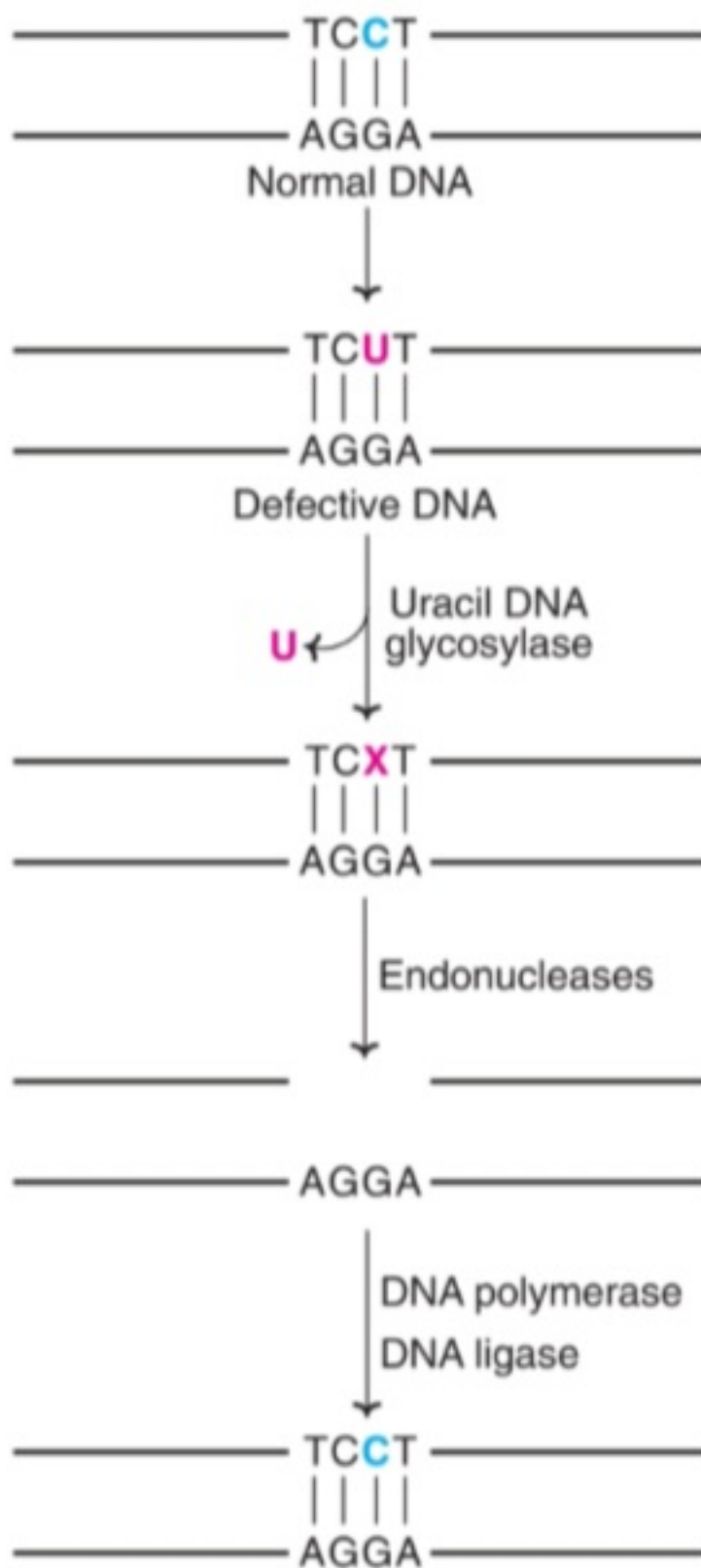


Fig. 24.16 : A diagrammatic representation of base excision-repair of DNA.

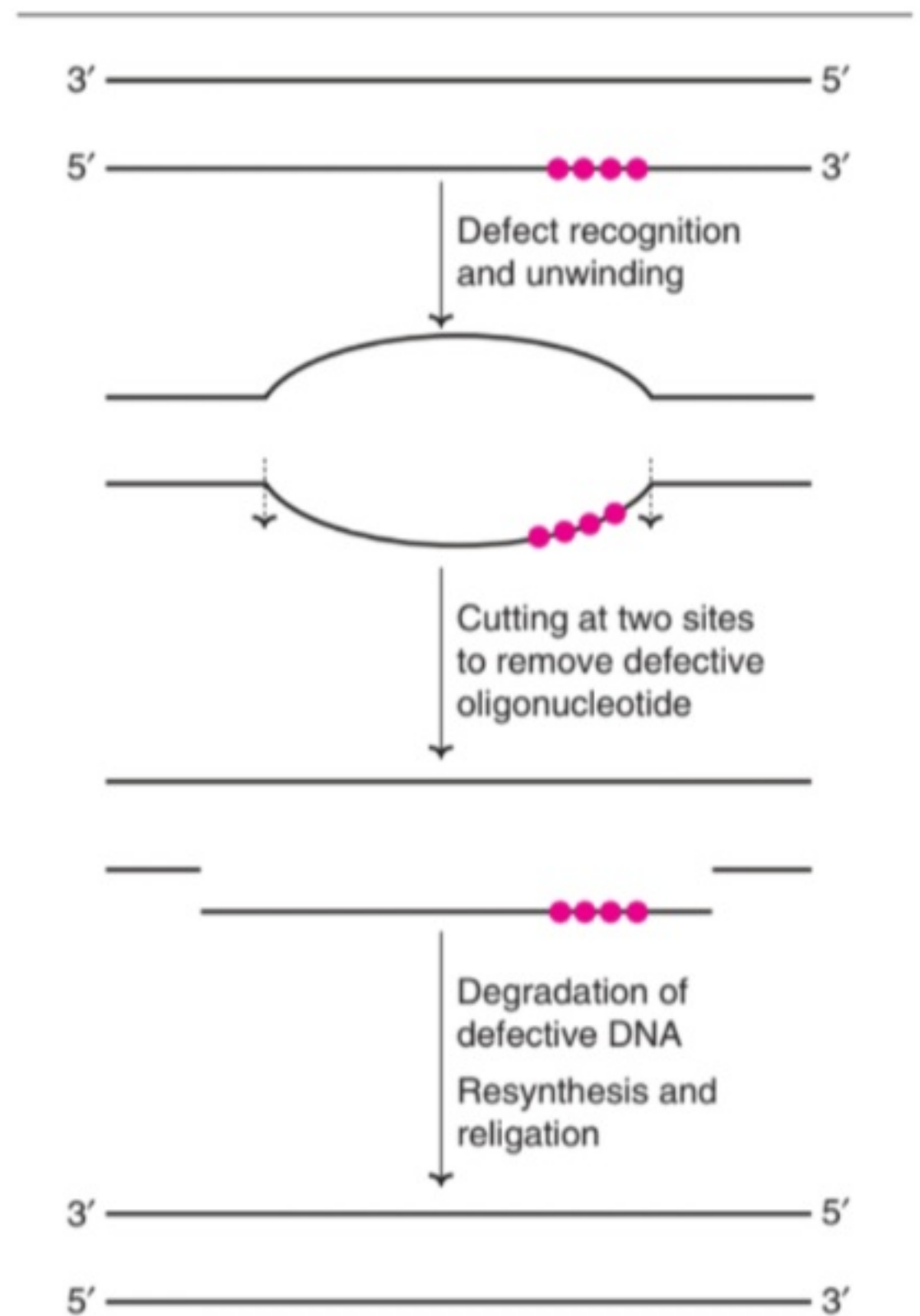


Fig. 24.17 : A diagrammatic representation of nucleotide excision-repair of DNA.

2) NUCLEOTIDE EXCISION-REPAIR:

The DNA damage due to ultraviolet light, ionizing radiation and other environmental factors often results in the modification of certain bases, strand breaks, cross-linkages etc. Nucleotide excision-repair is ideally suited for such large-scale defects in DNA. After the identification of the defective piece of the DNA, the DNA double helix is unwound to expose the damaged part. An **excision nuclease** (exinuclease) cuts the DNA on either side (upstream and downstream) of the damaged DNA. This defective piece is degraded. The gap created by the nucleotide excision is filled up by DNA polymerase which gets ligated by DNA ligase (Fig.24.17).

Xeroderma pigmentosum (XP) is a rare autosomal recessive disease. The affected patients are photosensitive and susceptible to skin cancers. It is now recognized that XP is due to a defect in the nucleotide excision repair of the damaged DNA.

3) MISMATCH REPAIR:

Despite high accuracy in replication, defects do occur when the DNA is copied. For instance, cytosine (instead of thymine) could be incorporated opposite to adenine. Mismatch repair corrects a single mismatch base pair e.g. C to A, instead of T to A.

The template strand of the DNA exists in a methylated form, while the newly synthesized strand is not methylated. This difference allows the recognition of the new strands. The enzyme GATC endonuclease cuts the strand at an adjacent methylated GATC sequence (Fig.24.18). This is followed by an exonuclease digestion of the defective strand, and thus its removal. A new DNA strand is now synthesized to replace the damaged one.

Hereditary nonpolyposis colon cancer (HNPCC) is one of the most common inherited cancers. This cancer is now linked with **faulty mismatch repair** of defective DNA.

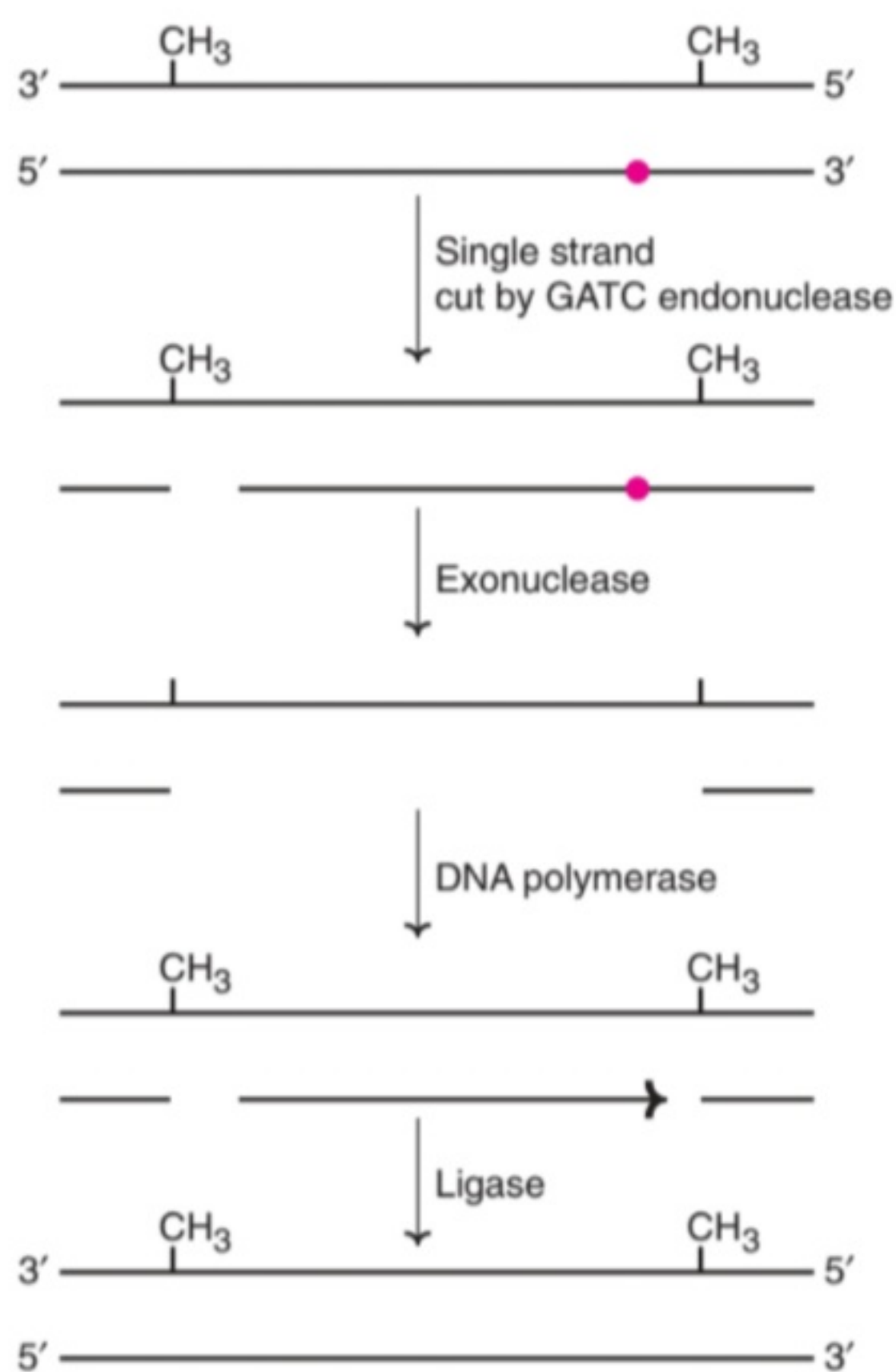


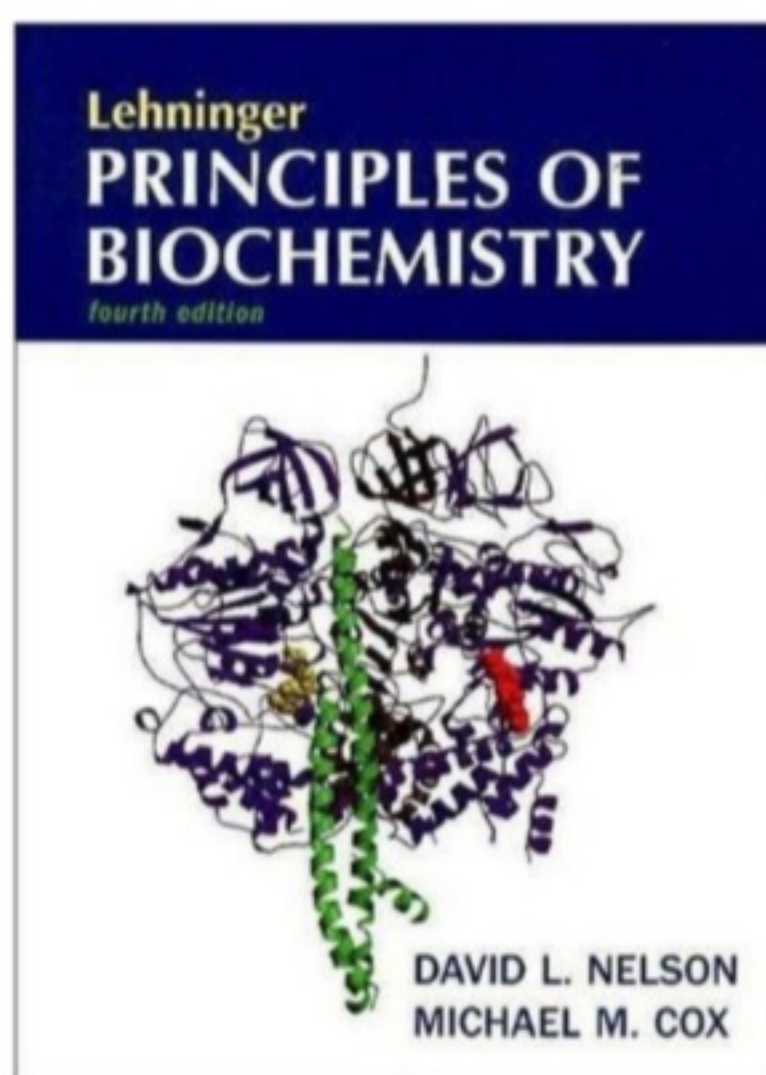
Fig. 24.18 : A diagrammatic representation of mismatch repair of DNA.

4) DOUBLE-STRAND BREAK REPAIR:

Double-strand breaks (DSBs) in DNA are dangerous. They result in genetic recombination which may lead to chromosomal translocation, broken chromosomes, and finally cell death. DSBs can be repaired by homologous recombination or non-homologous end joining. Homologous recombination occurs in yeasts while in mammals, non-homologous and joining dominates.



For more detailed information refer the below book.



Lehninger principles of biochemistry:

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